Multimodal Healing Phase Classification of Diabetic Foot Ulcer Using Jointly Optimized Fusion with Generative Augmentation

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***Abstract*— Diabetic foot ulcer (DFU) healing phase classification remains challenging due to wound progression complexity and current assessment limitations. This paper introduces FUSE4DFU, a multimodal fusion network for DFU healing phase classification integrating clinical metadata, RGB imaging, thermal mapping, and depth sensing. FUSE4DFU employs jointly optimized feature concatenation fusion, where all pipeline components (image backbones, metadata processing, fusion parameters, and ensemble weights) are co-optimized through Bayesian search rather than tuned independently. We validated FUSE4DFU using the Zivot dataset comprising 3,108 matched multimodal data points from 443 wound assessments across 233 patients. The best fusion combination achieved quadratic-weighted Cohen's kappa of 0.58 and accuracy of 73.3% in three-class healing phase classification (inflammatory, proliferative, remodeling), representing a statistically significant improvement over single-modality approaches ($p < 0.05$). An optimized ensemble of metadata-containing modality combinations achieved kappa of 0.65 and accuracy of 80.6% with balanced per-class performance. Generative augmentation using fine-tuned Stable Diffusion XL models improved minority class recall, with an inverted-U dose-response observed across augmentation probabilities. This framework enables objective, accessible wound assessment supporting phase-specific therapeutic interventions without laboratory diagnostics, advancing personalized diabetic foot care.**

***Index Terms*— Diabetic foot ulcer, multimodal classification, deep learning, feature fusion, wound healing, medical imaging, generative augmentation, Bayesian optimization, clinical decision support**

I. Introduction

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DIABETIC foot ulcers (DFUs) affect 6.3% of the diabetic population globally, with only 50% healing within one year despite optimal care [1], [2]. These complex wounds precede 85% of diabetes-related amputations [3] and consume 12-15% of diabetes-allocated healthcare resources [4]. With anticipated 25% increase in diabetes prevalence by 2030 [5], the disparity between DFU patients and specialized healthcare resources necessitates innovative assessment approaches.

Current clinical assessment relies on established classification systems including University of Texas [6], Wagner [7], and Wound, Ischemia, and Foot Infection (WIfI) [8] systems. While these excel in determining treatment urgency, they provide limited insight into dynamic cellular processes underlying tissue repair. Classification into inflammatory (I), proliferative (P), and remodeling (R) phases offers biological alignment with tissue repair processes, enabling phase-specific therapeutic interventions [9].

Phase-specific treatments demonstrate significant clinical value: anti-inflammatory agents during inflammatory phases [10], [11], growth factor therapies during proliferative phases [12], [13], and collagen modulators during remodeling phases [14]. However, implementation remains challenging due to absence of accessible, objective assessment methods independent of laboratory diagnostics [15], [16].

Recent computational approaches show promise but remain limited. Single-modality systems demonstrate high accuracy in binary tasks: red-green-blue (RGB) based methods achieve 93.4% accuracy [17] and thermal imaging shows 87.5% sensitivity for inflammatory detection [18], yet these fail to capture healing complexity of the other phases. Emerging multimodal approaches like DFU_VIRNet achieve F1-scores of 0.96 for RGB-thermal classification [19], but focus on binary paradigms rather than clinically relevant multi-class healing phase characterization.

Critical research gaps persist: (1) lack of systematic evidence for multimodal integration value in healing phase classification, (2) absence of comprehensive datasets preventing rigorous comparative analysis, and (3) limited frameworks addressing three-class temporal healing dynamics essential for clinical assessment.

This paper introduces FUSE4DFU (Fusion for Diabetic Foot Ulcer using 4 modalities), addressing these limitations through: (1) jointly optimized feature concatenation fusion

where image backbones, metadata classifiers, fusion parameters, and ensemble weights are co-optimized; (2) comprehensive evaluation of all 15 possible modality combinations establishing quantitative evidence for multimodal integration; (3) an ensemble framework combining predictions from metadata-containing modality combinations for balanced per-class performance; and (4) generative augmentation using fine-tuned Stable Diffusion XL (SDXL) models to address class imbalance in the minority remodeling phase.

Our contributions include:

1) Establishing generalizable insight into modality contribution hierarchy and performance patterns of clinical metadata, RGB, thermal, and depth modalities combine for healing phase discrimination relative to each other on a uniform dataset and architecture. This provides quantitative and reliable guidelines for resource-adaptive and clinically relevant design of multimodal DFU assessment systems.
2) Development of the FUSE4DFU architecture with jointly optimized feature concatenation fusion, where all pipeline components are co-optimized via Bayesian search rather than tuned independently, achieving accuracy of 80.6% in three-class classification.
3) Characterization of generative augmentation dose-response across three injection probabilities, identifying an optimal rate of 15% with diminishing returns beyond.

II. Methods

A. Experimental Framework and Ethical Approval

The Conjoint Health Research Ethics Board of the University of Calgary (#21-1052) and Research Ethics Board of the University Health Network (#21-5352) granted ethical approval for this multicenter research. Informed consent was obtained from all patients prior to data collection and participation in this study.

Our comprehensive evaluation framework analyzes effectiveness of different DFU assessment modalities in wound healing phase classification. As illustrated in Fig. 1, the pipeline assesses both single-modal and multimodal approaches, enabling direct comparison and reproducibility. The framework combines modalities through feature concatenation fusion, with an ensemble step that averages predictions from metadata-containing combinations for balanced per-class performance.

B. Dataset and Preprocessing

The Zivot dataset [20] contains wound assessments from 268 patients across multiple appointments, each capturing four modalities: clinical metadata, RGB imaging, thermal mapping, and depth sensing. A multimodal matching process aligned samples across modalities, retaining only those with complete records for all four. After screening out corrupted and mismatched images, the final dataset comprised 3,108 matched data points representing 443 unique wound assessments from 233 patients. Class distribution comprised inflammatory (I: 23.3%), proliferative (P: 63.2%), and remodeling (R: 13.1%) phases, reflecting natural clinical distributions with inherent class imbalance.

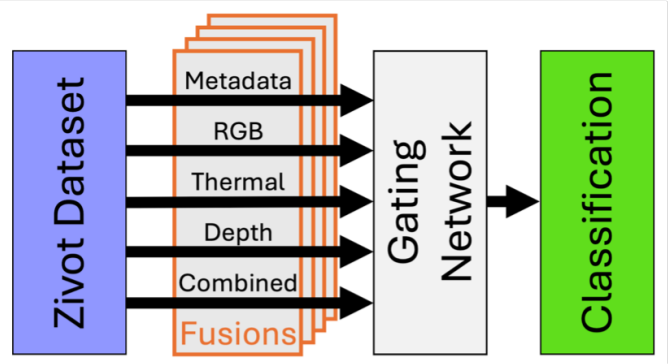

Fig. 1. Experimental Framework Overview. The multimodal classification system shows fusion layer formation containing modality combinations, processed through an ensemble framework using simple probability averaging for final classification.

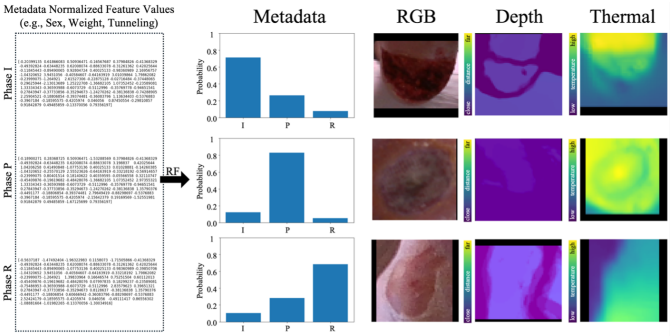

Fig. 2. Representative Examples of Multimodal Input Data. Visualization across modalities for each healing phase (Inflammatory: I, Proliferative: P, Remodeling: R), showing distinct characteristics captured by metadata, RGB images, depth maps, and thermal imaging. Metadata are processed through Random Forest producing calibrated three-class probability vectors.

Metadata processing uses 72 clinical features encompassing demographics, wound characteristics, temperature measurements, and treatment history, with missing values imputed using 5-nearest neighbors (less than 5% missing values). Treatment-related information was excluded to prevent healing phase data leakage [21]. A Random Forest (RF) classifier with 300 estimators, maximum depth of 10, produces calibrated three-class probability vectors via out-of-fold prediction, serving as the metadata representation for fusion.

For RGB, thermal, and depth images, preprocessing involved cropping to wound bounding box coordinates with modality-specific margins (field-of-view based scaling for depth, fixed 30-pixel expansion for thermal), and aspect-ratio preserving resize to 128×128 pixels.

Figure 2 shows representative examples of multimodal input data across healing phases, demonstrating distinct characteristics captured by each modality. All experiments employ five-fold patient-stratified cross-validation, ensuring no patient appears in both training and validation splits within any fold.

C. FUSE4DFU Architecture

1) Modality-Specific Processing: The metadata branch employs an RF classifier producing calibrated three-class probability vectors via out-of-fold prediction, as previously demonstrated for DFU healing phase classification [21]. The RF output (P_I, P_P, P_R) serves directly as the metadata representation in fusion, requiring zero trainable neural parameters for this branch.

All image modalities (RGB, thermal map, depth map) are processed through DenseNet121 [22] backbones pretrained on ImageNet [23]. DenseNet121 was selected over EfficientNet [24] and ResNet [25] variants through joint optimization experiments (Section II-D), where its dense connectivity pattern demonstrated superior transfer learning for false-color medical images. Each backbone is followed by a modality-specific projection head consisting of Dense, BatchNorm, and Dropout layers that reduce the feature dimensionality for fusion.

2) Feature Concatenation Fusion: FUSE4DFU employs feature concatenation fusion, where the RF probability vector (3 dimensions) is concatenated with projected image features from each included modality, then passed through a fully connected output layer producing three-class probabilities. This approach preserves modality-specific characteristics throughout processing while allowing the final layer to learn optimal weighting.

Training follows a two-stage protocol. In Stage 1, image backbones are frozen and only the projection heads and fusion output layer are trainable (up to 500 epochs with early stopping, learning rate 1.72×10^{-3}). In Stage 2, the top 5% of backbone layers are unfrozen at reduced learning rate (5×10^{-6}) for 30 additional epochs with batch normalization layers frozen.

The network trains using focal loss [26]:

$$\mathcal{L} = \alpha(1-p)^{\gamma}[-y\log(p)] \tag{1}$$

where p represents the predicted probability for the true class, $\gamma = 2.0$ is the focusing parameter, and α represents class-specific weights inversely proportional to class frequencies. This loss addresses class imbalance by down-weighting well-classified examples.

3) Ensemble Framework: The ensemble step combines softmax predictions from multiple modality combinations into a single prediction. A dedicated optimization audit evaluated 111 configurations across eight ensemble strategies: simple averaging, optimal weighted averaging [27], temperature scaling [28], logistic regression stacking [29], multilayer perceptron (MLP) stacking, gradient boosting stacking [30], rank-weighted averaging, and multi-head attention [31]. The optimal strategy was simple probability averaging of the eight modality combinations that include metadata. This parameter-free approach outperformed all learned ensemble methods, including the attention-based gating network which collapsed to majority-class prediction on some folds due to overfitting with limited training data.

4) Generative Augmentation: Generative augmentation employs a single conditional SDXL 1.0 model [32], fully fine-tuned on wound images from all three healing phases using phase-specific text prompts (e.g., "PHASE_I, diabetic foot ulcer, inflammatory phase wound"). The model generates synthetic RGB wound images at 512×512 resolution, which are resized to the training resolution of 128×128.

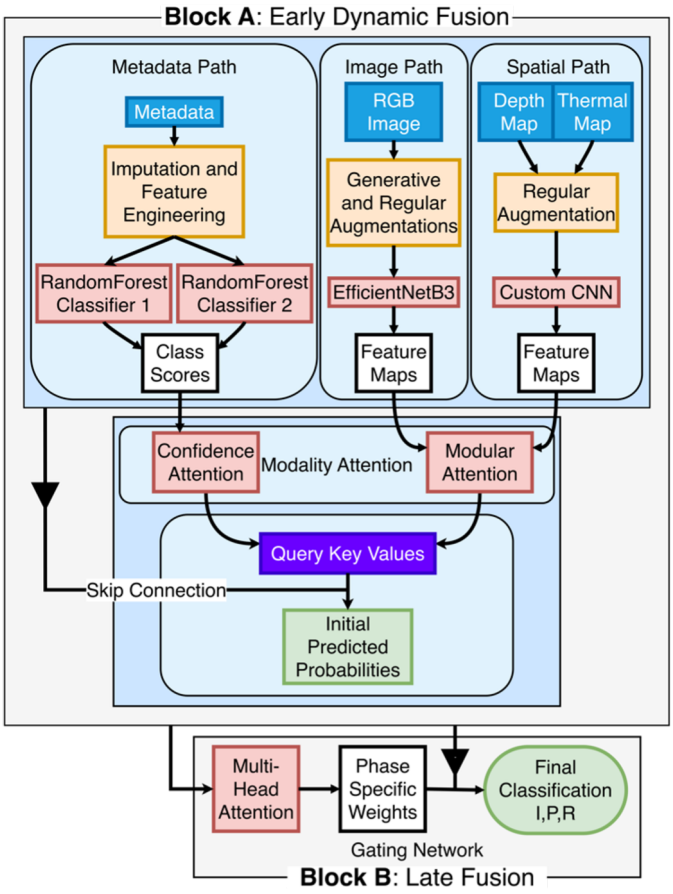

Fig. 3. FUSE4DFU Architecture. Each modality combination is processed through a fusion module (Block A) containing DenseNet121 image branches, Random Forest metadata branch, and feature concatenation fusion. The ensemble framework (Block B) combines predictions from multiple combinations via simple probability averaging.

To avoid GPU memory conflicts between the SDXL pipeline (PyTorch) and the training framework (TensorFlow), synthetic images are pre-generated to a disk cache before training begins. During training, cached images are randomly sampled and mixed into batches containing RGB modality data at a configurable probability. Generative augmentation is confined to RGB imagery, as thermal and depth modalities are measurement-based and require preservation of spatial registration and biophysical constraints. The augmentation probability was evaluated at 6%, 15%, and 25% to characterize the dose-response relationship.

Figure 3 illustrates the FUSE4DFU architecture. Each modality combination is processed through a fusion module (Block A) producing three-class predictions. When multiple combinations are evaluated, the ensemble framework (Block B) combines their predictions via simple averaging.

D. Joint Bayesian Optimization

A key methodological contribution is the joint optimization of all pipeline components simultaneously, rather than tuning each modality independently before fusion. Using Bayesian optimization [33] with Gaussian Process surrogate and Expected Improvement acquisition [34] (100 trials), the search co-optimized 23 hyperparameters spanning image backbones (DenseNet121 [22], EfficientNet [24], ResNet50V2 [25], MobileNetV3Large [35]), image resolution (32, 64, 128, 256), projection head architecture, RF parameters, fusion strategy, learning rates, training epochs, and loss function parameters. The top 10 configurations were validated with five-fold cross-validation to select the final architecture.

The rationale for joint optimization is that modality interactions during fusion create dependencies that sequential per-modality tuning [36] cannot capture. The resulting configuration and performance comparison are reported in Section III-B.

E. Evaluation Methodology

Evaluation employs five-fold cross-validation with patient-stratified splitting, preventing data leakage from longitudinal patient visits. Performance assessment uses multiple complementary metrics: accuracy, class-averaged F1-score, per-class F1-scores, receiver operating characteristic (ROC) curves, and quadratic-weighted Cohen's kappa coefficient [37], [38], appropriate for the ordinal nature of healing phases (inflammatory $\rightarrow$ proliferative $\rightarrow$ remodeling). Cohen's d effect size [**?**] quantifies the magnitude of inter-class separability in raw image features, with conventional thresholds of 0.2 (small), 0.5 (medium), and 0.8 (large). Statistical analyses include one-way ANOVA comparing performance across modality groups, linear regression quantifying the relationship between number of modalities and performance, and paired t-tests for focused comparisons [39]. Significance established at $p < 0.05$. Probability calibration was assessed via reliability diagrams and expected calibration error (ECE). The synthetic image generation quality was evaluated using Fréchet Inception Distance (FID) [40], Structural Similarity Index Measure (SSIM) [41], and Learned Perceptual Image Patch Similarity (LPIPS) [42].

III. Results

A. Generative Augmentation Performance

Quantitative evaluation of the fine-tuned SDXL model (Figure 4) at the selected checkpoint (epoch 35) showed overall FID of 220.13, SSIM of 0.87 $\pm$ 0.05, and LPIPS of 0.41 $\pm$ 0.10. Per-phase structural similarity was highest for remodeling (SSIM = 0.88, LPIPS = 0.37), followed by proliferative (SSIM = 0.85, LPIPS = 0.45) and inflammatory (SSIM = 0.84, LPIPS = 0.47). The elevated FID reflects the domain gap between medical wound images and ImageNet-pretrained Inception features used for FID computation, a known limitation for medical imaging [43]. To characterize the augmentation dose-response, three injection probabilities were evaluated: 6%, 15%, and 25%. The target modality combination (metadata + RGB + thermal) showed consistent improvement in minority class recall, with F1-R increasing from 0.52 (no augmentation) to 0.54 (6%), 0.55 (15%), and 0.55 (25%). The response followed an inverted-U pattern: at 25%, overall kappa declined from 0.59 to 0.50, indicating that excessive synthetic data injection causes overfitting to the fixed image cache. The optimal probability of 15% was selected, maximizing F1-R while maintaining overall performance.

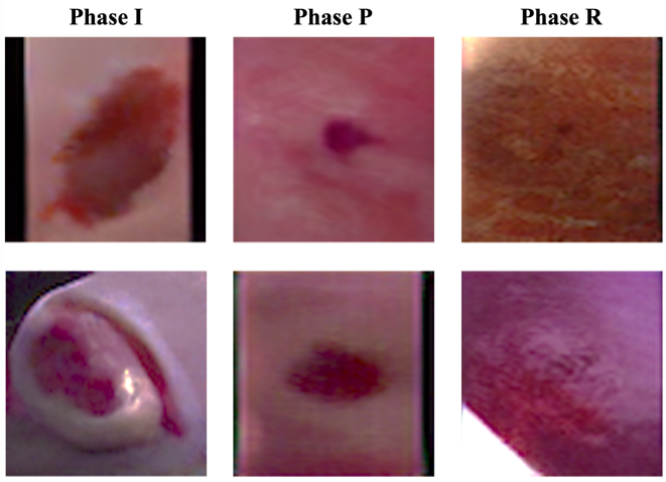

Fig. 4. Generated DFU Images Using SDXL. Examples of synthetically generated DFU images across healing phases (I, P, R), demonstrating generative augmentation capability in producing clinically relevant training samples at 512×512 resolution.

B. Single Modal and Multimodal Classification Results

Single-modal classification established important benchmarks. Thermal achieved the highest single-modality Cohen's kappa of 0.46 $\pm$ 0.12 with accuracy of 60.0%, followed by metadata (kappa 0.45 $\pm$ 0.17, accuracy 70.6%), RGB (kappa 0.45 $\pm$ 0.11, accuracy 62.7%), and depth (kappa 0.18 $\pm$ 0.07, accuracy 44.3%). The substantially lower depth performance is explained by analysis of the raw image data: depth map crops contain 15.9$\times$ fewer unique colors than RGB crops (median 78 vs 1,136), with 6.4% of depth crops being effectively a single solid color. Inter-class pixel intensity differences in depth maps are negligible (Cohen's d: I-vs-P = 0.23, I-vs-R = 0.16, P-vs-R = 0.08), indicating that wound surface topology does not vary meaningfully across healing phases. By contrast, thermal maps exhibit moderate class separation (Cohen's d: I-vs-R = 0.50, P-vs-R = 0.36, all $p < 0.0001$), consistent with the biological expectation that inflammatory wounds are warmer (increased blood flow) and remodeling wounds are cooler (reduced metabolic activity). ROC analysis (Figure 5) confirmed discriminative capability across modalities: the best fusion combination (metadata + RGB + thermal) achieved mean AUC of 0.87, with per-class AUC of 0.91 (inflammatory), 0.80 (proliferative), and 0.90 (remodeling). Single-modality AUCs ranged from 0.60 (depth) to 0.82 (metadata), demonstrating the additive value of multimodal integration.

Figure 6 shows phase-specific F1-scores across modality combinations. Despite class imbalance, metadata demonstrated

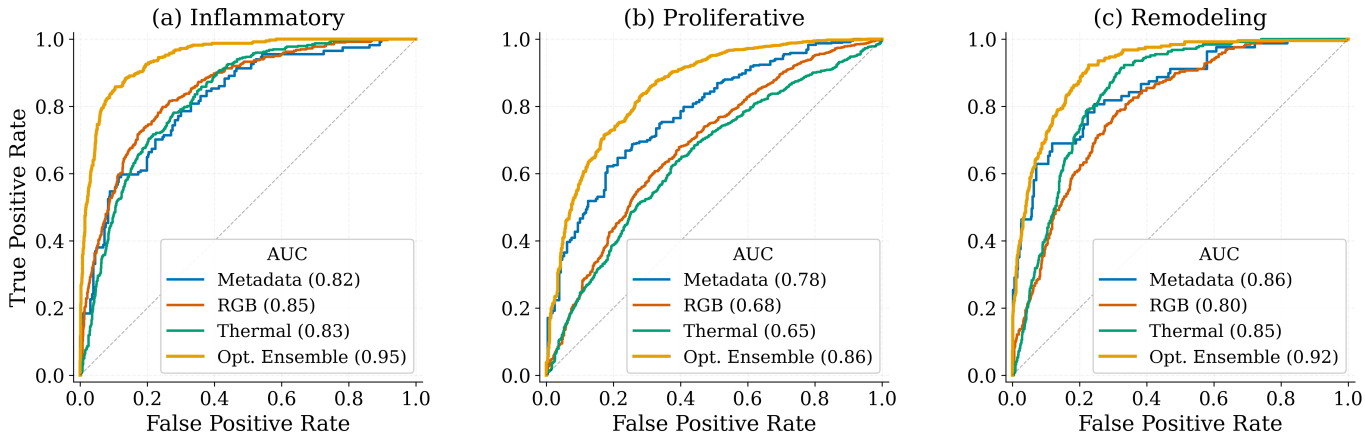

Fig. 5. ROC Curves by Healing Phase. Per-class receiver operating characteristic curves for the three single-modality baselines and the optimized ensemble (Opt. Ensemble) across five-fold cross-validation. AUC values in parentheses. The optimized ensemble achieves AUC of 0.95 (inflammatory), 0.86 (proliferative), and 0.92 (remodeling), outperforming all single modalities.

the strongest single-modality proliferative phase performance (F1-P: 0.79), while imaging modalities contributed complementary information for inflammatory phase discrimination.

Multimodal fusion demonstrated consistent improvements with increasing modality count. Cohen's kappa followed a clear trend: single 0.39, dual 0.50, triple 0.55, quaternary 0.58. One-way ANOVA confirmed significant performance differences across modality group sizes ($F = 5.56$, $p = 0.002$). Linear regression revealed a positive relationship between modality count and kappa (slope $= 0.07$, $R^2 = 0.40$, $p = 0.01$), indicating that each additional modality contributes approximately 0.07 kappa units. McNemar's test confirmed that the best fusion significantly outperformed metadata alone at the sample level ($\chi^2 = 4.56$, $p = 0.03$) and depth alone ($\chi^2 = 84.50$, $p < 0.001$).

The joint optimization discovered that DenseNet121 at 128×128 resolution with feature concatenation fusion outperforms the independently tuned configuration [36] (EfficientNetB0 at 256×256), improving five-fold mean kappa from 0.30 to 0.37, ultimately reaching 0.58 after complete pipeline optimization.

The best individual modality combination, metadata + RGB + thermal, achieved Cohen's kappa of 0.58 ± 0.12 and accuracy of 73.3%. Notably, the top four combinations (all including metadata and RGB) were within 0.01 kappa of each other, indicating that the core metadata + RGB signal accounts for most discriminative power, with thermal and depth providing marginal additional gains.

Figure 7 illustrates performance improvement with modality integration, showing incremental gains as additional modalities are incorporated into the fusion framework.

C. Modality Contribution Analysis

Analysis of standalone versus fusion performance revealed the relative contribution of each modality to the combined classification. Metadata and thermal achieved the highest single-modality kappas (0.45 and 0.46 respectively), while RGB provided the most complementary imaging information for fusion (kappa 0.45). Every fusion combination that included metadata outperformed metadata alone, confirming that image modalities provide additive discriminative value despite their comparable standalone performance. Depth maps showed the weakest contribution (kappa 0.18 alone), degrading some combinations they joined. This finding is supported by the raw image analysis: depth maps encode wound surface topology as viridis colormapped distance values, producing low color diversity (median 78 unique colors per crop) with minimal inter-phase variation. Wound surface geometry does not appear to differ systematically across healing phases, whereas wound color (RGB), temperature (thermal), and clinical history (metadata) each capture distinct phase-discriminative characteristics.

Sample-level agreement analysis across the three primary modalities (metadata, RGB, thermal) confirmed that metadata serves as the essential anchor modality, informing the ensemble design.

D. Ensemble Framework Results

The ensemble combines softmax predictions from all modality combinations that include metadata (8 of 15 total combinations). This parameter-free approach was selected from 111 configurations spanning eight strategies (Section II-C.3). Table I presents comprehensive performance analysis across modality combinations, comparing individual combination results with the ensemble.

The full ensemble (averaging all eight metadata-containing combinations) achieved accuracy of 78.5% (kappa 0.63) and balanced per-class performance among all configurations tested. Leave-one-out analysis of ensemble composition revealed that not all combinations contribute equally: metadata alone was the strongest contributor (+0.015 kappa when included), while some high-performing individual combinations (metadata + RGB + thermal) actually degraded ensemble performance (-0.016 kappa) due to overconfident predictions. An optimized four-combination subset improved ensemble kappa from 0.63 to 0.65 while maintaining accuracy of 80.6%. The selected combinations were:

- Metadata (standalone)
- Metadata + depth

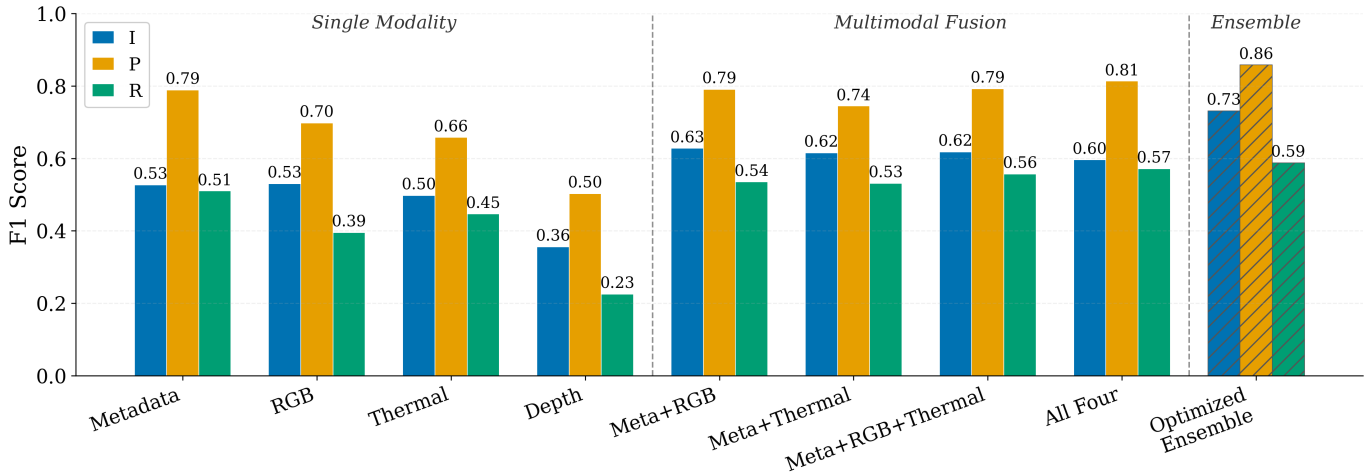

Fig. 6. Phase-Specific F1-Scores Across Modality Combinations. Comparative analysis of single-modality, fusion, and optimized ensemble (hatched) performance across healing phases in five-fold cross-validation. The ensemble achieves the highest proliferative F1 (0.86) and balanced per-class performance.

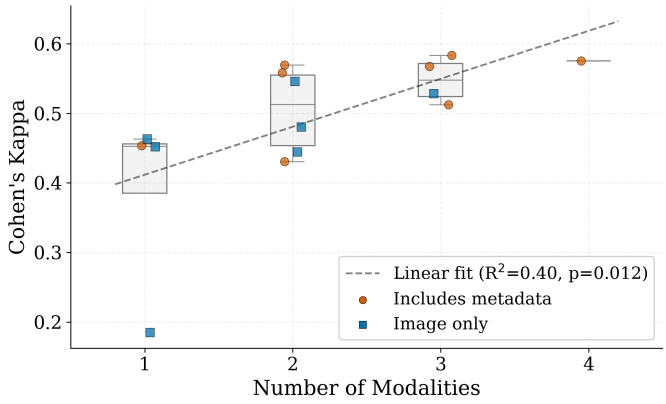

Fig. 7. Performance Progression with Modality Integration. Analysis showing incremental gains in Cohen's kappa as additional modalities integrate into the fusion framework across five-fold cross-validation.

- Metadata + thermal
- Metadata + RGB + depth

This finding indicates that ensemble diversity, not individual combination performance, determines ensemble effectiveness.

Figure 8a visualizes the ensemble composition: each bar represents one metadata-containing combination's individual kappa, with dark bars marking the four combinations selected for the optimized ensemble. The full ensemble averages all eight metadata-containing combinations, while the optimized ensemble uses only the four-combination subset identified through leave-one-out analysis. Notably, the optimized subset does not simply select the highest-performing individual combinations, confirming that prediction diversity drives ensemble effectiveness.

Figure 8b shows probability calibration via reliability diagrams [28], where proximity to the diagonal indicates alignment between predicted confidence and actual accuracy. The best fusion combination (Meta+RGB+Thermal) achieved ECE of 0.08, indicating well-calibrated predictions, while metadata alone showed ECE of 0.16. The optimized ensemble exhibited higher ECE (0.27), as probability averaging across models compresses confidence values, reducing calibration quality. Despite this, the ensemble achieves the highest accuracy (80.6%) and most balanced per-class F1, suggesting that the accuracy gains outweigh the calibration cost for clinical decision support.

TABLE I

PERFORMANCE ANALYSIS ACROSS KEY MODALITY COMBINATIONS. I = INFLAMMATORY, P = PROLIFERATIVE, R = REMODELING, META = METADATA, RGB = RED-GREEN-BLUE, STD = STANDARD DEVIATION, κ = COHEN'S KAPPA. COMPLETE RESULTS FOR ALL 15 COMBINATIONS ARE PROVIDED IN APPENDIX A, TABLE II.

Modality Combination	**F1 (%)**			**Accuracy**	**Cohen's**
	I	**P**	**R**	**±std (%)**	**κ ±std**
Single modalities					
Metadata	53	79	51	71 ±07	0.45 ±0.17
RGB	53	70	39	63 ±04	0.45 ±0.11
Thermal	50	66	45	60 ±05	0.46 ±0.12
Depth	36	50	23	44 ±12	0.18 ±0.07
Top fusion combinations					
Meta + RGB + Thermal	62	79	56	73 ±05	0.58 ±0.12
Meta + RGB	63	79	54	73 ±04	0.57 ±0.10
Meta + RGB + Depth	61	80	55	74 ±03	0.57 ±0.09
Meta + Thermal	62	74	53	69 ±07	0.56 ±0.11
All Four	60	81	57	75 ±04	0.58 ±0.10
Ensemble					
Full (8 combos)	**68**	**84**	**60**	**79 ±04**	**0.63 ±0.11**
Optimized (4 combos)	**73**	**86**	**59**	**81 ±05**	**0.65 ±0.12**

IV. DISCUSSION

A. Clinical Significance and Innovation

FUSE4DFU represents a significant advancement in automated DFU assessment, with the optimized ensemble achieving Cohen's kappa of 0.65 and accuracy of 80.6% in healing phase classification without laboratory diagnostics. This performance approaches practical clinical decision support

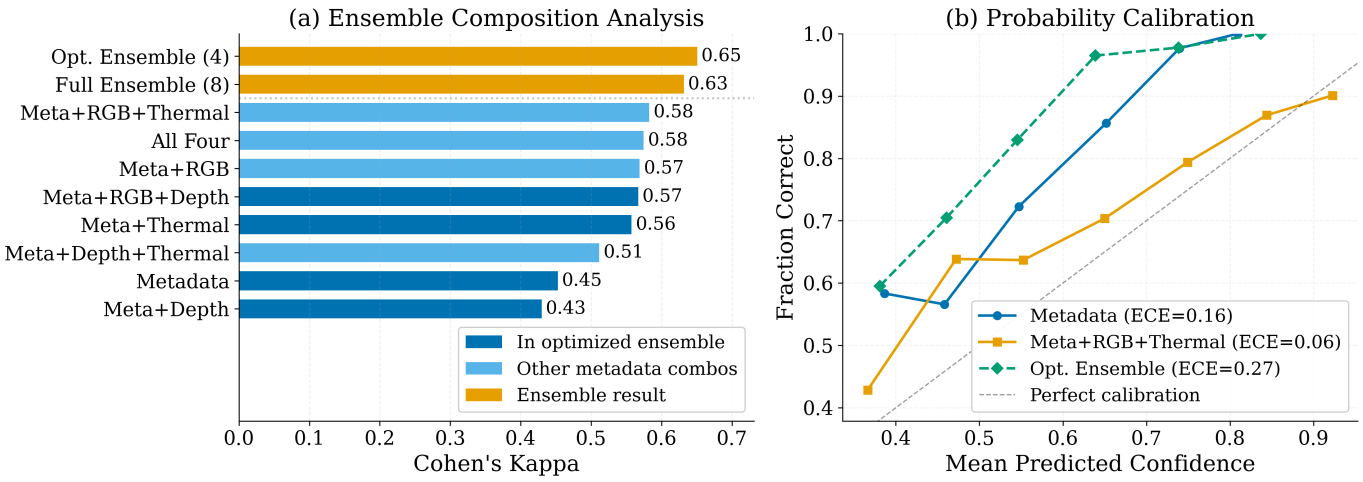

Fig. 8. Ensemble Analysis. (a) Ensemble composition showing Cohen's kappa for each metadata-containing combination individually, with the full (8-combination) and optimized (4-combination) ensembles. Dark bars indicate combinations selected for the optimized ensemble. (b) Probability calibration reliability diagram with expected calibration error (ECE) for key configurations.

requirements while maintaining accessibility across diverse healthcare settings. Multimodal fusion improved mean kappa from 0.39 (single-modality average) to 0.58 (best fusion combination), with ensemble averaging further increasing kappa to 0.65, demonstrating substantial value of multimodal integration in capturing complex wound healing progression.

The modality contribution analysis revealed a clear implementation hierarchy: metadata and thermal provide the strongest standalone signals (kappa 0.45 and 0.46 respectively), consistent with prior findings on the predictive power of clinical metadata for DFU healing trajectory [21], [44], while RGB imaging offers the most complementary information (every metadata + RGB combination outperforms metadata alone). This hierarchy provides evidence-based guidance for resource-adaptive clinical implementation, from basic electronic health record integration to comprehensive multimodal assessment systems. Importantly, thermal mapping contributes to balanced per-class performance, particularly for the minority remodeling phase (F1-R: 0.53 for metadata + thermal vs 0.54 for metadata + RGB).

From a clinical perspective, these findings offer insight into which physical characteristics differentiate healing phases. Metadata (patient demographics, wound scores, temperature readings) captures systemic factors that broadly predict healing trajectory, while RGB imaging captures wound surface morphology (color, texture, tissue composition) that distinguishes inflammatory from proliferative phases. Thermal mapping captures subsurface vascular activity that is particularly informative for identifying the remodeling phase, where reduced metabolic activity produces distinct thermal signatures. Depth sensing alone carries minimal phase-discriminative information (kappa 0.18), suggesting that wound topology is not a reliable indicator of healing phase in isolation. These observations inform both clinical assessment priorities and future sensor development for wound monitoring devices.

Existing clinical classification systems such as the University of Texas, Wagner, and WIfI focus on wound severity, infection risk, and amputation probability, providing essential treatment urgency assessment. FUSE4DFU complements these systems by adding temporal healing phase characterization, which these classification systems do not address. Where WIfI determines *what* intervention is needed based on current wound state, healing phase classification determines *when* to adjust treatment based on the biological stage of tissue repair. Integrating both approaches could enable clinicians to combine severity assessment with phase-aware treatment selection, for example adjusting growth factor therapies during the proliferative phase or identifying stalled inflammatory responses that require intervention.

B. Comparison with Existing Approaches

Direct comparison reveals novel contributions versus existing systems. While Al-Garaawi et al. [45] reported F1-scores of 0.95-0.99 for binary RGB-thermal classification, and Reyes-Luévano et al. [19] achieved 0.96 for binary classification, these approaches do not address temporal healing complexity. Our three-class healing phase discrimination provides substantially higher clinical value, directly informing phase-specific therapeutic interventions.

The jointly optimized fusion framework achieved Cohen's kappa of 0.58 for the best individual combination, with optimized ensemble kappa of 0.65 and accuracy of 80.6%, representing a novel contribution to multimodal fusion strategies for medical imaging. A key finding is that feature concatenation fusion with jointly optimized components outperforms more complex attention-based approaches when training data is limited (443 samples). This aligns with established machine learning principles favoring simpler models under data constraints [43].

It is important to note that the absolute performance values reported in this study are influenced by the available dataset size and should be interpreted primarily in relation to each other, rather than as definitive benchmarks. The central contribution of this work is the systematic characterization of how modalities combine and contribute on a uniform dataset with

consistent sample counts across all modalities. This controlled setup enables fair comparison between modality combinations, revealing the relative value of each sensor type. Absolute performance can be expected to improve with larger datasets, but the relative modality hierarchy and fusion dynamics observed here provide generalizable insights for multimodal wound assessment system design.

C. Joint Optimization and Architectural Insights

The joint Bayesian optimization revealed several findings that contradict conventional assumptions in medical image analysis. First, DenseNet121 [22] outperformed EfficientNet [24] for both RGB and thermal modalities, despite EfficientNet's widespread adoption in medical imaging. DenseNet's dense connectivity and feature reuse may provide superior transfer learning for false-color medical images that differ substantially from ImageNet's natural image domain. Second, simple probability averaging of metadata-containing combinations outperformed all 111 tested ensemble strategies including multi-head attention, MLP stacking, and gradient boosting. The attention-based gating network collapsed to majority-class prediction on two of five folds, confirming that complex ensemble architectures are prone to overfitting with limited training data.

The calibration analysis reveals an inherent tradeoff in ensemble averaging: while individual fusion models achieve low calibration error (ECE 0.08), the ensemble's probability averaging compresses confidence values, increasing ECE to 0.27 despite improving classification accuracy from 73.3% to 80.6%. In clinical deployment, this implies that the ensemble's predicted class is more often correct, but its associated confidence score is less reliable as an uncertainty estimate. For applications requiring both accurate predictions and trustworthy confidence scores, post-hoc calibration methods such as temperature scaling [28] could be applied to the ensemble output.

D. Generative Augmentation and Technical Innovation

Implementation of SDXL-based generative augmentation addresses the fundamental challenge of class imbalance in medical datasets. The dose-response characterization across three injection probabilities (6%, 15%, 25%) revealed an inverted-U relationship: F1-R for the minority remodeling class improved from 0.52 (no augmentation) to 0.55 (15%), then declined at 25% due to overfitting to the fixed synthetic image cache. This finding provides practical guidance for calibrating augmentation intensity, which has received limited systematic investigation in medical imaging [43]. FID scores confirmed preservation of phase-specific characteristics in the generated images.

This work represents among the first applications of SDXL-based synthetic image generation specifically for DFU wound classification, building on our earlier work applying diffusion models to DFU image synthesis [46] and vision-language models for DFU assessment [47], [48].

E. Resource Adaptation and Implementation

FUSE4DFU's modular architecture enables effective implementation across diverse healthcare settings. In our previous work [49], we demonstrated that a metadata-driven approach, utilizing a larger dataset, achieves 69% accuracy and provides fundamental triage capabilities. In the present study, requiring all four modalities for each sample reduces the available dataset size, yet metadata alone still achieves 70.6% accuracy. Following the observed increasing performance trend with additional modalities, the integration of RGB imaging with metadata should be prioritized (kappa improvement from 0.45 to 0.57), followed by thermal imaging for improved minority class performance. Sites supporting all four modalities can implement the full FUSE4DFU architecture with ensemble framework, achieving the highest overall accuracy (80.6%).

F. Limitations and Future Directions

Several limitations warrant acknowledgment. Class imbalance (I: 23%, P: 63%, R: 13%), while reflecting clinical reality, presents analytical challenges particularly for the remodeling phase, potentially affecting generalizability. The single-site dataset may not capture demographic diversity across healthcare systems, requiring multi-institutional validation.

Imaging equipment requirements for multimodal acquisition may represent barriers for primary care adoption, though costs are reasonable for specialized wound care centers. Each image preprocessing step isolated single wound areas, limiting capability for simultaneous multiple ulceration analysis present in complex clinical scenarios.

Future directions include longitudinal validation studies enabling healing trajectory prediction, multi-site validation across diverse clinical settings, and clinician-in-the-loop studies evaluating optimal human-artificial intelligence collaboration strategies. Technical advances could include domain-specific pretraining on wound image corpora, expanded generative datasets with diversity-aware sampling, and physics-informed generative models preserving measurement characteristics across modalities.

V. Conclusion

This paper introduced FUSE4DFU, a jointly optimized multimodal fusion framework for three-class DFU healing phase classification. Through systematic evaluation of all 15 modality combinations on a uniform dataset, this work established that clinical metadata, RGB imaging, and thermal mapping provide complementary discriminative value, with joint Bayesian optimization of all pipeline components improving kappa from 0.30 to 0.58 compared to sequential tuning. An optimized ensemble of metadata-containing combinations achieved kappa of 0.65 and accuracy of 80.6%, and generative augmentation at 15% injection probability improved minority class recall following an inverted-U dose-response. Experimental validation on 3,108 matched multimodal data points from 233 patients confirmed statistically significant improvements ($p < 0.05$). The modular architecture supports resource-adaptive deployment from metadata-only triage to comprehensive multimodal assessment, enabling objective

wound evaluation supporting phase-specific therapeutic interventions without laboratory diagnostics.

Data and Code Availability

The code used in this study is publicly available through the GitHub repository "DFUMultiClassification" at `https://github.com/rezabasiri/DFUMultiClassification`. The resulted model weights are also provided in this repository for deployment and evaluation on publicly available single modality datasets such as DFU2020 [50] or Plantar Thermogram Database [51]. The four-modality training dataset cannot be shared publicly at this point because the release of data was not included in the ethics approval. The modality analyses and provision of the FUSE4DFU method with a detailed description of features collected from patients presented in previous works [20], [21] stands as an independent contribution.

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APPENDIX I
COMPLETE MODALITY COMBINATION RESULTS

TABLE II
COMPLETE PERFORMANCE ANALYSIS FOR ALL 15 MODALITY COMBINATIONS (FIVE-FOLD PATIENT-STRATIFIED CROSS-VALIDATION WITH 15% GENERATIVE AUGMENTATION). I = INFLAMMATORY, P = PROLIFERATIVE, R = REMODELING, META = METADATA, RGB = RED-GREEN-BLUE, ACC = ACCURACY, STD = STANDARD DEVIATION, κ = COHEN'S KAPPA.

Combination	**F1-I**	**F1-P**	**F1-R**	**Acc±std**	**κ±std**
Single modalities					
Metadata	0.53	0.79	0.51	71±07	0.45±0.17
RGB	0.53	0.70	0.39	63±04	0.45±0.11
Thermal	0.50	0.66	0.45	60±05	0.46±0.12
Depth	0.36	0.50	0.23	44±12	0.18±0.07
Dual modalities					
Meta + RGB	0.63	0.79	0.54	73±04	0.57±0.10
Meta + Thermal	0.62	0.74	0.53	69±07	0.56±0.11
Meta + Depth	0.52	0.75	0.45	67±04	0.43±0.10
RGB + Thermal	0.59	0.76	0.51	70±04	0.55±0.12
RGB + Depth	0.54	0.75	0.45	68±03	0.48±0.10
Depth + Thermal	0.49	0.66	0.41	59±05	0.44±0.12
Triple modalities					
Meta+RGB+Thermal	0.62	0.79	0.56	73±05	0.58±0.12
Meta+RGB+Depth	0.61	0.80	0.55	74±03	0.57±0.09
Meta+Depth+Thermal	0.55	0.74	0.50	68±03	0.51±0.09
RGB+Depth+Thermal	0.58	0.76	0.48	69±02	0.53±0.10
All four modalities					
Meta+RGB+Depth+Thermal	0.60	0.81	0.57	75±04	0.58±0.10